INCH-POUND

MIL-C-45574B(AR)
20 February 1991
SUPERSEDING
MIL-C-45574A(AR)
9 September 1987
and
MIL-B-45588A(WC)
14 January 1972
Notice 1
12 March 1986

MILITARY SPECIFICATION

CANNON, 152MM GUN-LAUNCHER: M81E1 AND M162
(Including Breech Mechanisms)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification supports the acquisition of two types of 152MM Gun-Launcer Cannon, each consisting of a breech mechanism assembly and a cannon tube assembly. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

1.1.1 Supersession. MIL-B-45588A, Notice 1, Breech Mechanism Asseblies for Cannon, 152MM Gun Launcher, M81E1 and XM162E1 has been cancelled and the contents thereof included in this specification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SS, Watervliet, N.Y. 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1025

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1.2 Classification. Cannon shall be of the following types as specified (see 6.2):

M81E1 - Cannon, 152MM Gun - Launcher. For use on the M551 Armored Reconnaissance Airborne Assault Vehicle.

M162 - Cannon, 152MM Gun-Launcher. For use on the M60A2 Tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the is issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-13931 - Cannon, General Specification for

MIL-P-14232 - Parts, Equipment, and Tools for Army Material, Packaging of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance, and Disposition of

MIL-L-46150 - Lubricant, Weapons, Semi-fluid (High Load-carrying Capacity)

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1189 - Standard Department of Defense Bar Code Symbology

MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing, and Marking

MIL-STD-1949 - Inspection, Magnetic Particle

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

EPL 7986980 - Breech Mechanism Assembly

EPL 11578068 - Engineering Parts List for Cannon, 152MM Gun-Launcher:

EPL 11578135 - Breech Mechanism Assembly

EPL11578233 - Engineering Parts List for Cannon, 152 Gun-Launcher:

DL 11578068 - List of Inspection Equipment Numbers for: Cannon, Gun

DL 11578233 - List of Inspection Equipment Numbers for: Cannon, Gun

PUBLICATIONS

Master LQAP - Master Index of Quality Assurance Provisions for Cannon. 11578068 152MM Gun-Launcher: M81E1

Master LQAP - Master Index of Quality Assurance Provisions for Cannon. 11578233 152MM Gun-Launcher: M162

PDS P11578068 - Packaging Data Sheet, Cannon, 152MM Gun-Launcher: M81E1

PDS P11578233 - Packaging Data Sheet, Cannon, 152MM Gun-Launcher: M162

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

2.2 Non-Government publications. The following documents(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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American National Standards Institute (ANSI)

ANSI B46.1 - Surface Texture

ANSI Y14.5 - Dimensioning and Tolerancing

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York City, New York 10018.)

American Society for Testing and Materials (ASTM)

ASTM D3951 - Commercial Packaging, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicble laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Cannon, breech mechanism assemblies, and component parts shall conform to the applicable drawings and specifications listed in applicable Cannon Engineering Parts List EPL 11578068 (M81E1) or EPL 11578233 (M162), and applicable breech mechanism assembly Engineering Parts List EPL 7986980 (M81E1) or EPL 11578135 (M162), the requirements of this document, the applicable paragraphs of MIL-C-13931, as incorporated in this document, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.4 and 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer (see 6.2).

3.3 Breech mechanism assembly.

3.3.1 Breech mechanism assembly for M81E1 cannon. The breech mechanism for the M81E1 cannon shall be assembled. In accordance with the requirements of drawing 7986980.

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3.3.2 Breech mechanism assembly for M162 cannon. The breech mechanism for the M162 cannon shall be assembled in accordance with the requirements of drawing 11578135.

3.4 Autofrettage. Breech chambers, as well as tubes, shall be autofrettaged in accordance with the requirements of MIL-C-13931 for tubes.

3.5 Lubrication. Apply lubricant MIL-L-46150 to all movable and sliding surfaces except ball screw threads before functional testing.

3.6 Operation. The cannon shall operate in any attitude within 35 degrees elevation, 23 degrees depression, and 15 degrees left or right cant and any ambient temperature between -65F and +140F.

3.6.1 Electrical Operation of Breech Mechanism Assembly.

3.6.1.1 Power required. The breech mechanism assembly shall operate with applied drive-motor terminal voltages from 18 to 30 volts DC. At 24 volts ± 1 volt and at an ambient temperature of 75 + 5F, with the exception of a transient surge at the initiation of an opening or a closing phase, the current shall not exceed 40 amperes. In any 10 complete cycles, the increase in current value shall not exceed five amperes for the conditions specified in this paragraph and 3.6.

3.6.1.2 Breech mechanism assembly operating time. The breech mechanism operating time at any attitude specified in 3.6, and at any voltage from 18 to 30 volts DC, shall not exceed 3.5 seconds for either an opening or a closing phase of a cycle and shall not exceed 6.5 seconds for one complete cycle. In addition, at 24 volts + 1 volt DC, and at an ambient temperature of 75 + 5F, the operating time shall not exceed 2.8 seconds for either an opening phase or a closing phase and shall not exceed 5 seconds for one complete cycle.

3.6.2 Manual operation. The maximum input torque at the hand crank to operate the breech mechanism at zero degree (0^{0}) elevation and between fifteen degrees (15°) left to fifteen degree (15°) right cant shall not exceed 300 pound-inches at ambient temperatures between -65F and +140F.

3.6.3 Obturator, cap-detent, and firing-probe seals. The obturator seal shall be readily removable. The obturator, cap-detent, and firing-probe seals shall prevent the escape of hot gases and powder residue into the turret area.

3.6.3.1 Seal area surfaces. Damage shall be prevented while cleaning preserving, and handling parts used in the obturator seal areas. These surfaces shall be protected during machining and handling.

3.6.4 Breech chamber closure. The breech mechanism assembly whall prevent movement of the breech when fully closed. When fully opened, the breech mechanism assembly shall prevent the breech chamber from closing under shock and vibration of vehicle operation.

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3.6.5 Scavenger hardware.

3.6.5.1 Scavenger hardware M81E1. The scavenger hardware shall operate with gas bursts of 0.8 ± 0.1 second duration at a pressure of 300 + 20 psi.

3.6.5.2 Scavenger hardware (M162). The scavenger hardware plunger and chamber valves shall operate with gas bursts of 0.35 + 0.01 second duration at a pressure of 1180 + 20 psi.

3.7 Proof testing The cannon shall be capable of withstanding a chamber pressure of 43,400 ± 1,500 psi

3.7.1 Material soundness. Before and after proof firing, all parts shall be free from cracks and fractures and shall comply with any special soundness requirements specified on the applicable drawing.

3.7.2 Bore and breech chamber enlargement. Enlargement of the tube bore as a result of proof firing shall not exced 0.003 inch after a maximum of twenty (20) proof rounds. No enlargement of the breech chamber is permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain confrormance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified on the drawings or applicable paragraphs in the specification.

4.4 First article inspection. Unless otherwise specified in the contract (see6 2), first article cannon assemblies, breech mechanism assemblies, and components thereof shall be submitted for inspection in accordance with the contract and applicable Quality Assurance Provisions (QAPs). The first article assemblies and components thereof shall be representative of the production processes to be used during quantity production. QAPs are listed in Master LQAP 11578068 and Master LQAP 11578233.

4.4.1 First article sample.

4.4.1.1 Sam le size. The sample size shall be as specified in the contract (see 6.2) and each assembly and component in the sample shall be randomly 6.2) selected from the first month's production.

4.4.2 First article inspection. The first article sample shall be subjected to all quality conformance inspections and all examinations and tests described in this specification to assure conformance with the requirements of this specification. Any required additional inspections, ncluding First Article - Initial Production Testing, shall be specified in the contract (see 6.2).

a. Failure of the first article sample to meet the requirements as evidenced by the results of the examinations and tests shall be cause for rejection (see 6.2). The contractor shall determine the cause of failure and shall take corrective action in current and subsequent production. Resubmission and retest shall be at the option of the Government (see 6.2).

b. Disposition of an acceptable first article sample shall be in accordance with the provisions of the contract (see 6.2)-

4.5 Quality conformance inspection.

4.5.1 Inspection conformance. Gun-launchers shall be inspected for conformance to this specification, to MIL-C-13931, and to QAPs listed in Master LQAP 11578068 (M81E1) or Master LQAP 11578233 (M162). In addition, breech mechanism assemblies shall be inspected in accordance with the requirements for the applicable breech mechanism assembly drawing. Quality conformance inspections are tabulated in Table I.

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TABLE I. Quality conformance inspection.

Characteristic	Requirement	Examination or Test
1. Firing circuit		
a. resistance	4.5.7.4	4.5.7.4
b. insulation	4.5.7.5	4.5.7.5
2. Harness assembly		
a. insulation	4.5.7.6	4.5.7.6
3. Lubrication	3.5	Visual
4. Breech mechanism operation	3.6.1	4.5.7
a. Manual operation	3.6.2	Torque measurement
b. Electrical operation	3.6.1.1 3.6.1.2	4.5.7.2
(1) 18-volt setting	3.6.1.1 3.6.1.2	4.5.7.2.1
(2) 30-volt setting	3.6.1.1 3.6.1.2	4.5.7.2.2
(3) 24-volt setting	3.6.1.1 3.6.1.2	4.5.7.2.3
(4) current limit	3.6.1.1	4.5.7.2.4
(5) limit switch setting	4.5.7.1	4.5.7.1
c. Obturator, cap-detent, and firing probe seals	3.6.3 3.6.3.1	Visual
d. Breech chamber closure	3.6.4	4.5.9.2.6
5. Scavenger	3.6.5	4.5.7.7
6. Environmental testing	3.6	4.5.8
a. Breech assembly cycling-manual	3.6	4.5.8.1
b. Breech assembly cycling-electrical	3.6	4.5.8.2

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TABLE I. Quality conformance inspection. (continued)

Characteristic	Requirement	Examination or Test
7. Proving ground testing		
a. Proof firing	3.7	4.5.9, 4.5.9.1
b. Material soundness	3.7.1	4.5.9.2.5
(1) Tube	3.7.1	4.5.4, 4.4.9.2.1 4.5.9.2.2
(2) Coupling and chamber	3.7.1	4.5.5$_{s}$ 4.5.9.2.1
(3) Carrier	3.7.1	4.5.6, 4.5.9.2.1
c. Obturator, cap-detent, and firing probe seals	3.6.3 3.6.3.1	4.5.9.2.3
d. Bore and breech chamber enlargement	3.7	4.5.9.2.4
e. Breech chamber closure	3.6.4	4.5.9:2.6
f. Scavenger	3.6.5	4.5.7*7
8. Packaging	5.	4.5.10.1

4.5.1.1 Final examination. Final examination of each gun-launcher shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract (see 6.2).

4.5.1.2 Inspection approval stare . The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Inspection equipment. Data lists containing equipment used in the inspection of parts, sub-assemblies, and assemblies are listed in DL 11578060 (M81E1) Or DL 11578233 (M162).

4.5.2.1 Acquisition, maintenance, and disposition. Unless otherwise specified, responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed in applicable item QAPs (Quality Assurance Provisions), and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

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4.5.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.3 Protective coating. Tests of protective coatings shall be conducted in accordance with the applicable drawings and specifications. A minimum of one sample specimen for each applicable test shall be selected from the lot being inspected. The lot shall be subject to rejection if any specimen fails in any of the tests.

4.5.4 Tubes. All external and internal surfaces of each tube shall be examine magnetic particle inspection in accordance with MIL-STD-1949.

4.5.5 Coupling and chamber. All external and internal surfaces of each coupling and each chamber shall be examined by magnetic particle inspection in accordance with MIL-STD-1949.

4.5.6 Carrier. All external and internal surfaces of each carrier shall be examine magnetic particle inspection in accordance with MIL-STD-1949.

4.5.7 Function tests. Each gun-launcher shall be subjected to the following tests. These tests may be run on the breech mechanisms as a separate assembly if so designated by the procuring agency. In this event, tests 4.5.7.2.3 (position "A" only) and 4.5.7.4 shall be re-run on the completed gun-launcher. Cannon failing to meet requirements shall be rejected.

4.5.7.1 Limit switch setting. The circuit across pins K and N of the main disconnect shall be interrupted 90 degrees minimum prior to crank reaching the mechanical stop in opening direction. The circuit across pins K and L of - the main disconnect shall be interrupted 60 *degrees* minimum prior to crank reaching the mechanical stop in the closing direction.

NOTES: a. Satisfactory manual actuation must be accomplished prior to any electrical actuation.

b. A cycle, with respect to operation of the breech assembly, is defined as the opening followed by the closing of the breech chamber.

4.5.7.2 Electrical actuation.

4.5.7.2.1 18 volt setting. At an ambient temperature of 75 + 5F with an impressed DC voltage of 18 volts +2, -0 volts applied at the launcher harness connection, cycle the breech assembly three times in each position “A” thru "c". Record the motor current, voltage, and time required for each-cycle. The time for each partial cycle shall not exceed 3.5 seconds either “to open” or “to close” the breech chamber. The total actual time for each complete" cycle shall not exceed 6.5 seconds (this does not include cycle delay time).

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NOTES :

a. Electrical cycle actuation rate. During all electrical cycle actuation, a maximum of six complete cycles per minute shall not be exceeded.

b. Delay - After the unit is energized to open breech, a time delay of 2.0 seconds shall exist prior to energizing the unit to close breech. This same minimum time delay shall exist prior to each “opening” and each “closing” portion of any cycle.

c. Breech assembly positions (position accuracy ±2°):

Position “A” - level

Position “B” - 35° elevation and 15° right can't

Position “C” - 23° depression and 15° left can't

4.5.7.2.2 30 volt setting. At an ambient termperature of 75 + 5F, with an impressed DC voltage of 30 volts +0, -1 volt applied at the launcher harness connection, cycle the breech assembly three times in each position “A” thru “c”. This voltage setting operation is considered satisfactory when no evidence of malfunction is detected. The time for each partial cycle shall not exceed 3.5 seconds either “to open” or “to close” the breech chamber. The total actual time for each complete cycle shall not exceed 6.5 seconds (this does not include cycle delay time).

4.5.7.2.3 24 volt setting. At an ambient temperature of 75 + 5F, with an impressed DC Voltage 4 volts + 1 volt applied at the launcher harness D connection, cycle the breech assembly five times in each position “A” thru “c”. Record the motor current, voltage, and time required for each cycle. The time for each partial cycle shall not exceed 2.8 seconds either “to open” or “to close” the breech chamber. The total actual time for each complete cycle shall not exceed 5.0 seconds (this does not include cycle delay time). The current shall not exceed 40 amperes under any condition of breech attitude with the exception of a transient surge at cycle initiation.

4.5.7.2.4 Limit in current value increase. The increase in current value shall be limited to a 5 ampere maximum rise within any 10 cycles of electrical operation of the breech mechanism when tested under conditions specified in 3.6.1.1.

4.5.7.3 Ejection test. Ejection shall be checked with an empty missile aft-cap. Ejection shall be considered acceptable if the empty cap is fully moved onto the loading tray where it may be brushed off by hand.

4.5.7.4 Firing circuit resistance test. The breech assembly harness resistance for the firing probe circuit shall be checked; the resistance shall not exceed 250 milliohms measured between harness input terminals “G” and “P” with harness connected to probe.

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NOTE : The resistance shall be checked while using a suitable dummy connector to provide a closed circuit across the firing probe. The resistance value limit shall include firing probe assembly resistance only, without resistance of dummy connector or its attachment.

4.5.7.5 Firing circuit megger test. With the harness assembly installed to all breech electrical connections, apply 500 volts DC for one (1) to five (5) seconds to the harness input "Pins A and C" with all other leads and the breech mechanism simultaneously grounded. The firing circuit shall show a minimum resistance of 100 megohms to ground with the breech closed. Accuracy of this test equipment shall be within 5 percent.

4.5.7.6 Harness assembly megger test. With the harness assembly removed from the breech assembly apply 500 volts DC for one (1) ti five (5) tp all pins of the harness input connector except "Pin P" with all the connector shells, mounting-bracket and mounting-block grounded. The harness assembly shall show a minimum resistance of 100 megohms on all wires to each other and to ground.

4.5.7.7 Scavenger hardware test (M162). Warning: Observe appropriate safetY precautions while testing with compressed air.

a. Open the breech mechanism so that the breech chamber is fully opened. With gas bursts of specified pressure and duration, operate both plunger valves a minimum of three cycles. The plunger valve should extend into the interior of the cannon on application of gas pressure and completely retract on removal of gas pressure. Verify by audible and visual means, the operation of the plunger valves.

b. Close the breech mechanism so that the breech chamber is fully closed and locked. With gas bvurst of specified pressure and duration, operate both chamber valves a minimum of three times. Verify by audible means the operation of the chamber valves.

4.5.8 Environmental test. The first production gun-launcher and one from each succeeding 100 units shall be tested electrically and manually at temperatures of +140F and -65F in position "A" and shall operate without special aids.

4.5.8.1 Breech assembly cycling - manual. The breech assembly shall be operated manually five cycles prior to electrical operation to check for evidence of malfunction. The maximum input torque at the hand crank to operate the breech mechanism at position "A", between 15^0 left and 15^0 right cant, shall not exceed 300 pound-inches at ambient temperatures of +140F and -65Fo

4.5.8.2 Breech assembly cycling - electrical. With 24 volts ± 1 volt DC applied at the launcher harness connections, cycle the breech assembly five times. Record motor current, voltage, and time required for each cycle.

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4.5.9 Proving ground inspection. Cannon, repair tubes, repair couplings, and repair chambers shall be subjected to proving ground inspection in accordance with International Test Operations Procedures at a test agency designated by USATECOM. The inspection shall be performed by Government personnel.

4.5.9.1 Proof firing
chamber shall be proof fired with one round capable of producing a pressure of 43,400 ± 1500 psi.

4.5.9.2 Inspection after proof firing_.

4.5.9.2.1 Magnetic particle inspection. Examine all external surfaces of the tube and all surfaces of the carrier, chamber, and coupling by magnetic particle inspection conducted in accordance with MIL-STD-1949. Acceptance criteria shall be as stated on the applicable drawing.

4.5.9.2.2 Bore inspection. Examine the bore surfaces of the tube with a borescope for evidence of cracks and other defects.

4.5.9.2.3 Gas leakage inspection. Examine the obturator, cap-detent, and firing-probe seals, and adjacent areas of the mating parts for evidence of gas leakage" during proof firing. In addition, disassemble the obturator seal to ascertain that it is readily removable after firing.

4.5.9.2.4 Bore and breech chamber enlargement inspection. Measure the bore breech chamber diameters with a star gage, air gage, or tother suitable instrument to determine compliance with 3.7.2. Measurements shall be performed after firing not more than 20 proof rounds.

4.5.9.2.5 After proof firing, the cannon shall not be disassembled below the sub-assembly level. Staked parts shall not be disassebled.

4.5.9.2.6 Breech chamber closure. The breech mechanism when assembled to the vehicle, shall be operated under specific conditions, The breech mechanism shall be inspected to determine compliance with 3.6.4.

4.5.9.3 Proof marking. Proof acceptance marking shall be located on the tube near the serial number

4.5.10 Inspection of packaging. The classification of, and inspection method for, defects for determining compliance with the requirements for packaging are listed in Table II.

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TABLE II. Classifiication of and inspection method for defects.

Category	Defect	Inspection method
Major		
101	Illegible or incorrect marking	Visual
102	Improper level of preservation, packaging or packing	Visual
103	Inadequate cleaning or drying	Visual
104	Improper preservation application	Visual
105	Improper or inadequate blocking and bracing	Visual/Manual
106	Improper closure or strapping of containers	Visual/Manual
Minor		
201	Workmanship	Visual

5. PACKAGING

5.1 Cannon - Levels A, B and C. Cleaning, drying, preservation, unit packaging and packing of cannon shall be in accordance with the requirements of Packaging Data Sheet (PDS) P11578068, P11578233, and MIL-P-14232 for the level of protection specified in the contract (see 6.2).

5.2 Repair parts - Level A, B and C. Cleaning, drying, preservation, unit packaging and packing of repair parts shall be in accordance with the applicable repair part PDS and MIL-P-14232 or the applicable repair part SPI and MIL-STD-2073-1 for the level of protection specified in the contract (see 6.2). When Level C requirements are not shown on the PDS/SPI, Level C packaging shall be in accordance with MIL-STD-1190.

5.2.1 Level X - Industrial packaging. Cleaning, drying, preservation, unit packaging and packing of repair parts shall be in accordance with ASTM 03951 when industrial packaging is specified in the contract (see 6.2).

5.2.1.1 Quantity in Unit Package (QUP). When industrial packaging of repair parts is specified in the contract, the QUP shall be the same as specififed in the contract (see 6.2).

5.3 Marking of packages.

5.3.1 Levels A, B and C. Marking of the unit package, intermediate packages. and exterior shipping containers shall be in accordance with MIL-STD-129. When specified (see 6.2), bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

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5.3.2 Level X - industrial. Marking of the unit package, intermediate packages, and exterior shipping containers shall be in accordance with ASTM 03951. When specified (see 6.2), bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cannon specified herein is intended to provide a vehicle mounted means for firing both conventional rounds of ammunition and rockets in close combat support of troops. The 152MM M81E1 gun launcher is used on the M551 Armored Reconnaissance Airborne Assault Vehicle. The 152MM M162 gun launcher is used on the M60A2 tank.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. a. Title, number and date of the specification.

b. b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.2).

d. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

e. All inspection and test data that is required.

f. At whose expense a retest may be performed.

g. Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract including, but not limited to, First Article - Initial Production Testing (FA-IPT). FA-IPT tests shall be conducted on cannon, critical assemblies, and components when:

(1) A new contractor is utilized.

(2) A time lapse of one year has occurred since the item was last produced by the contractor.

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(3) Significant changes have been made in a manufacturing process or material.

(4) A design change is made.

Note: The determination to the significance of the action with regard to the requirement for testing shall be made by the Product Assurance element of the procuring activity.

h. The serial numbers for the cannon and components.

i. List of inspection equipment, responsibility for acquisition. maintenance, and disposition thereof, if other than as specified (see 4.5.2).

j. Availability of inspection equipment from the Government.

k. Extent of contractor's responsibility for Government furnished and for contractor-required final inspection equipment.

l. Applicable acceptance test procedures.

m. When a proof firing sampling plan is authorized, the proof firing sampling plan parameters: initial consecutive quantity (i), and sampling frequency (f) (see 4.5.9.1).

n. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and repair parts (see 5.2).

o. When applicable, bar code marking requirements (see 5.3).

p. When warranted, the application of MIL-Q-9858 *or* MIL-I-45208, as appropriate.

q. Unless otherwise specified, the application of MIL-I-45607 and MIL-STD-45662.

r. Replacement parts should be provided to Government Proving Grounds and Watervliet Test Facilities to support the replacement of parts required after testing.

s. Reliability and Maintainability (RAM) tests and test methods shall be specified when required as part of FA-IPT.

6.3 Consideration of Data requirements. The following data requirements should be considered when this specification is appliedd h a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the

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data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-T-4902	First Article Inspection Report	

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the Procuring Contracting Officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.4. The Procuring Contracting Officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Product Assurance Directorate at Watervliet Arsenal.

6.6 Subject term (key word) listing.

Cannon, 152MM: M81E1
Cannon, 152MM: M162
Gun-Launcher: 152MM: M81E1
Gun-Launcher: 152MM: M162
Armored Reconnaissance Airborne Assault Vehicle: M551
Tank: M60A2
Breech Mechanism, Cannon, 152MM: M81E1
Breech Mechanism, Cannon, 152MM: M162

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6.7 Drawings. Drawings listed in Section 2 of this specifcation under the heading "U.S. Army Armament Research, Development and Engineering Center" (ARDEC) may also include drawings prepared by, and identified as Watervliet Arsenal, Ordnance Corps, Weapons Comman, or U.S. Army Armament Research and Development Command, U.S. Army Armament, Munitions and Chemical Command, etc. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army-AR

Preparing activity:

Army-AR

(Project 1025-A173)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-45574B (AR)	20 February 1990

3. DOCUMENT TITLE

CANNON, 152MM GUN-LAUNCHER: M81E1 AND M162

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC ATTN: SMCAR-CCB-SS		
c. ADDRESS (Include Zip Code) Watervliet, NY 12189-4050	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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